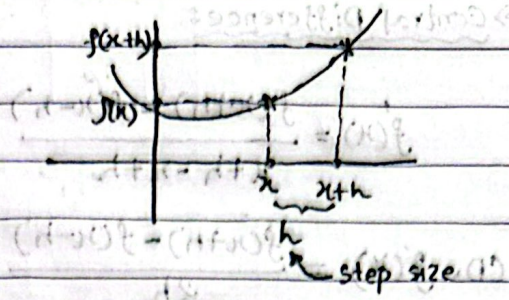


⇒ Forward Difference:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{f(x+h) - f(x)}{x+h - x}$$

$$\text{FD, } f'(x) = \frac{f(x+h) - f(x)}{h}$$



Example: $f(x) = x^2 + 10x$. Find $f'(x)$ using FD at $x=2$, $h=0.1$

$$\Rightarrow \text{Actual, } f'(x) = 2x + 10$$

$$f'(2) = 14$$

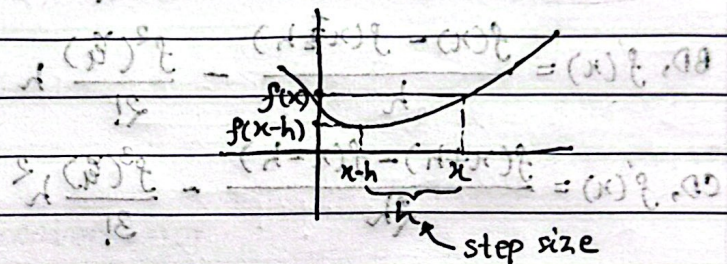
$$\text{Using FD, } f'(x) = \frac{f(x+h) - f(x)}{h} = \frac{f(2+0.1) - f(2)}{0.1} = \frac{2.1^2 + 10 \cdot 2.1 - (2^2 + 10 \cdot 2)}{0.1}$$

$$f'(2) = 14.1$$

⇒ Backward Difference:

$$\text{BD, } f'(x) = \frac{f(x) - f(x-h)}{x - (x-h)}$$

$$f'(x) = \frac{f(x) - f(x-h)}{h}$$

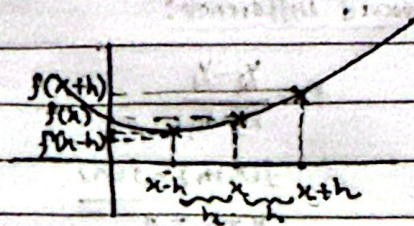


Q last example using BD,

$$\begin{aligned} f'(2) &= \frac{f(2) - f(2-0.1)}{0.1} \\ &= \frac{2^2 + 10 \cdot 2 - (1.9^2 + 10 \cdot 1.9)}{0.1} \\ &= 13.9 \end{aligned}$$

Central Difference:

$$f'(x) = \frac{f(x+h) - f(x-h)}{x+h - x-h}$$



$$CD, f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

Same example using CD,

$$\Rightarrow f'(2) = \frac{f(2+0.1) - f(2-0.1)}{2 \times 0.1}$$

$$= 14$$

(*) $f'(x) = \text{Approximate Value} + \text{Truncation Error}$

$$FD, f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{f''(\xi)}{2!} h$$

$$BD, f'(x) = \frac{f(x) - f(x-h)}{h} - \frac{f''(\xi)}{2!} h$$

$$CD, f'(x) = \frac{f(x+h) - f(x-h)}{2h} - \frac{f''(\xi)}{3!} h^2$$

So, we get,

Forward & Backward

Error $\propto h$

Central

Error $\propto h^2$

Q Example: $f(x) = \ln(x)$, $x=2$, $h = [1, 0.1, 0.01]$

find $f'(2)$ and truncation error

$\Rightarrow$ Using FD,

$$f'(2) = \frac{f(2+h) - f(2)}{h} = \frac{\ln(2+h) - \ln(2)}{h}$$

Actual, $f'(x) = \frac{1}{x}$, $f(2) = \frac{1}{2} = 0.5$

h	FD, $f'(x)$	Truncation Error
1	0.405465	$0.5 - 0.405465 = 0.09453$
0.1	0.487902	0.012098
0.01	0.498754	0.001246
0.001	0.499875	0.000124

here, as $h \downarrow$ error $\downarrow$

Q Example: $f(x) = 2x^2 - e^x$, $x=2$, $h = [0.1, 0.01, 0.001]$. find $f'(2)$ and Truncation Error.

$\Rightarrow$ CD, $f'(2) = \frac{f(x+h) - f(x-h)}{2h}$

Actual, $f'(x) = 4x - e^x$

$f(2) = 0.6109439$

h	CD, $f'(x)$	Truncation Error
0.1	0.5986226	0.0123213
0.01	0.6108207	0.0001232
0.001	0.6109427	0.0000012

Q Example:

x_0	4	4.1	4.2	4.3	4.4
$f(x_0)$	16	18	20	21	22

$$f'(4.2) = ?$$

$$h = 0.1$$

→ Using forward & backward.

$$D_{h, \text{forward}} = \frac{f(4.2+0.1) - f(4.2)}{0.1} = \frac{21-20}{0.1} = 10$$

$$D_{h, \text{backward}} = 20$$

Q Example: $v(t) = 2000 \ln \left(\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right) - 9.8t$

Calculate the value of acceleration $a(16) = [v'(16)]$ using FD, BD, CD.

given, $\Delta t = 2 \text{ s}$

Actual,

$$\text{FD, } a(16) = v'(16) = \frac{v(16+2) - v(16)}{2} = 30.474$$

$$\text{BD, } a(16) = v'(16) = 28.9145$$

$$\text{CD, } a(16) = v'(16) = 29.69437$$

* Error = Truncation Error + Rounding Error.

Considering CD.

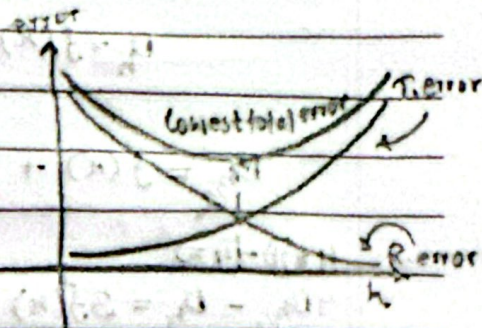
$$\text{Error} = \frac{f'''(\xi)}{3!} h^2 + \epsilon_M \cdot \frac{f(x+h) - f(x-h)}{2h}$$

$h \downarrow$ error $\downarrow$

$h \uparrow$ error $\uparrow$

$h \downarrow$ error $\uparrow$

$h \uparrow$ error $\downarrow$



Truncation Error:

Central: $\frac{f'''(\xi)}{3!} h^2 \Rightarrow$ Error is order of $h^2 = O(h^2)$

Forward, Backward: $\frac{f''(\xi)}{2!} h \Rightarrow$ Error is order of $h = O(h)$

→ Richardson Extrapolation: (for Central only)

- Define error in a better way.

- Reduces the order of error.

$h < 1$

$$\text{CD, } \rightarrow D_h = \frac{f(x+h) - f(x-h)}{2h}$$

Expand using Taylor series

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \frac{f^{(4)}(x)}{4!}h^4 + \frac{f^{(5)}(x)}{5!}h^5 + O(h^6)$$

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2!}h^2 - \frac{f'''(x)}{3!}h^3 + \frac{f^{(4)}(x)}{4!}h^4 - \frac{f^{(5)}(x)}{5!}h^5 + O(h^6)$$

$$f(x+h) - f(x-h) = 2f'(x)h + 2\frac{f'''(x)}{3!}h^3 + 2\frac{f^{(5)}(x)}{5!}h^5 + O(h^7)$$

$$D_h = \underbrace{f'(x)}_{\text{Actual Derivation}} + \underbrace{\frac{f'''(x)}{3!}h^2 + \frac{f^{(5)}(x)}{5!}h^4 + O(h^6)}_{\text{Truncation Error}}$$

We will try to remove $(h^2, h^4, \dots)$ terms one by one:

Lets starts with (h^2)

$$D_h = f'(x) + \frac{f''(x)}{2!} h^2 + \frac{f'''(x)}{3!} h^3 + O(h^4) \quad \text{--- (1)}$$

$$D_{h/2} = f'(x) + \frac{f''(x)}{2!} \frac{h^2}{4} + \frac{f'''(x)}{3!} \frac{h^3}{8} + O(h^4) \quad \text{--- (2)}$$

$$4 \times (2) - (1) \Rightarrow$$

$$4D_{h/2} - D_h = 3f'(x) + \left(\frac{1}{4} - 1\right) \frac{f''(x)}{2!} h^2 + O(h^4)$$

$$\Rightarrow \frac{4D_{h/2} - D_h}{3} = f'(x) - \frac{1}{4} \frac{f''(x)}{2!} h^2 + O(h^4) = D_h^{(1)} \quad \text{1 term is removed}$$

$$\therefore D_h^{(1)} = \frac{4D_{h/2} - D_h}{3}$$

What about h^4 term / $D_h^{(2)}$???

$$\Rightarrow D_h^{(1)} = f'(x) - \frac{1}{4} \frac{f''(x)}{2!} h^2 + O(h^4) \quad \text{--- (1)}$$

$$D_{h/2}^{(1)} = f'(x) - \frac{1}{4} \frac{f''(x)}{2!} \frac{h^2}{4} + O(h^4) \quad \text{--- (2)}$$

$$16 \times (2) - (1) \Rightarrow$$

$$16D_{h/2}^{(1)} - D_h^{(1)} = 15f'(x) + O(h^4)$$

$$\frac{16D_{h/2}^{(1)} - D_h^{(1)}}{15} = f'(x) + O(h^4) = D_h^{(2)}$$

$$\therefore D_h^{(2)} = \frac{16D_{h/2}^{(1)} - D_h^{(1)}}{15}$$

$$D_h^{(1)} = \frac{4D_{h/2} - D_h}{3} \Rightarrow \text{removes } \frac{1}{3} \text{ term}$$

$$D_h^{(2)} = \frac{16D_{h/2} - D_h}{15} \Rightarrow \text{removes } \frac{1}{15} \text{ term}$$

$$D_h^t = \frac{2^P D_{h/2}^{t-1} - D_h^{t-1}}{2^P - 1}$$

Ex: $D_h^2 = ?$

$$\Rightarrow \frac{4D_{h/2} - D_h}{3} = \frac{16D_{h/2} - D_h}{15}$$

$D_h^3 = ?$

$$\Rightarrow t=3, P=6 \quad D_h^3 = \frac{2^6 D_{h/2}^2 - D_h^2}{2^6 - 1} = \frac{64D_{h/2}^2 - D_h^2}{63}$$

Ex: $f(x) = e^x \sin(x)$, find $D_h^{(1)}$ using Richardson Extrapolation at $x=1$ for $h=0.5$.

$$\Rightarrow t=1, P=2 \quad D_h^1 = \frac{4D_{h/2} - D_h}{3}$$

$$D_h = \frac{f(x+h) - f(x-h)}{2h} = \frac{e^{1.5} \sin(1.5) - e^{0.5} \sin(0.5)}{2 \times 0.5} = 3.68$$

$$D_{h/2} = \frac{e^{1.25} \sin(1.25) - e^{0.75} \sin(0.75)}{2 \times 0.75} = 3.7385$$

$$D_h^1 = \frac{4 \times 3.7385 - 3.68}{3} = 3.757 \quad (\text{Ans})$$

Q1 Example: $f(x) = e^{2x} + 3x$. Find $f'(2)$ using Richardson Extrapolation
for $h=1.2$ & 0.6 .

$$\Rightarrow f'(x) = D_h = \frac{f(2+1.2) - f(2-1.2)}{2 \times 1.2} = \frac{e^{2(3.2)} + 3 \times 3.2 - (e^{2(0.8)} + 3 \times 0.8)}{2 \times 1.2}$$

$$= 251.705$$

$$D_{\frac{h}{2}} = \frac{f(2+0.6) - f(2-0.6)}{2 \times 0.6} = \frac{e^{2(2.6)} + 3 \times 2.6 - (e^{2(1.4)} + 3 \times 1.4)}{2 \times 0.6}$$

$$= 140.356$$

$$D_h^{(2)} = \frac{2^2 D_{\frac{h}{2}} - D_h}{2^2 - 1} = \frac{4 \times 140.356 - 251.705}{3}$$

$$= 103.23 \quad (\text{Ans})$$

Q2 Example: $f'(1) = ?$ $h=0.4$ Using: $D_h^{(2)}$

x	$f(x)$
0.6	0.707178
0.8	0.8559892
0.9	0.925863
1	0.984007
1.1	1.033743
1.2	1.074575
1.4	1.127986

$$\Rightarrow t=2 \quad p=4 \quad D_h^{(2)} = \frac{2^4 D_{\frac{h}{2}} - D_h}{2^4 - 1} = \frac{16 D_{\frac{h}{2}} - D_h}{15}$$

$$D_h = \frac{4 D_{\frac{h}{2}} - D_h}{3}$$

$$D_{\frac{h}{2}} = \frac{4 D_{\frac{h}{4}} - D_{\frac{h}{2}}}{3}$$

$$82.2 - 2885.0 \times 10^{-4} = 1.0$$

$$D_h = \frac{f(1+0.4) - f(1-0.4)}{2 \times 0.4} = \frac{f(1.4) - f(0.6)}{2 \times 0.4} = \frac{1.127986 - 0.707178}{2 \times 0.4}$$

$$= 0.52601$$

$$h_{1/2} = 0.2; \quad h_{1/4} = 0.1$$

$$D_{h_{1/2}} = \frac{f(1.2) - f(0.8)}{2 \times 0.2} = \frac{1.074575 - 0.8552892}{2 \times 0.2}$$

$$= 0.5464645$$

$$D_{h_{1/4}} = \frac{f(1.1) - f(0.9)}{2 \times 0.1} = 0.5394$$

$$D_h^1 = \frac{4 \times 0.5464645 - 0.52601}{3}$$

$$D_h^1 = \frac{4 \times 0.5394 - 0.5464645}{3}$$

$$= 0.5532826667$$

$$= 0.5370451667$$

$$D_h^{(2)} = \frac{16 \times 0.5370451667 - 0.5532826667}{15}$$

$$= (1) - (2) \times \frac{C}{\Delta t}$$

$$= 0.5359626667$$

(Ans)

$$D_h = \frac{f(1.0) + \frac{1}{2} \left(\frac{C}{\Delta t} \right) \frac{(C)^2}{12} + (C)^2}{1 - \frac{C}{\Delta t}} = D_h - \frac{D_h \frac{C}{\Delta t}}{1 - \frac{C}{\Delta t}}$$

$$\frac{D_h - \frac{D_h \frac{C}{\Delta t}}{1 - \frac{C}{\Delta t}}}{1 - \frac{C}{\Delta t}} = D_h \therefore$$

Ex Example: Deduce an expression for D_h^1 from D_h by replacing h with $(\frac{4h}{3})$ with Richardson Extrapolation method.

$$\Rightarrow D_h = \frac{f(x+h) - f(x-h)}{2h}$$

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f^{(3)}(x)}{3!}h^3 + \frac{f^{(4)}(x)}{4!}h^4 + \frac{f^{(5)}(x)}{5!}h^5 + O(h^6)$$

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2!}h^2 - \frac{f^{(3)}(x)}{3!}h^3 + \frac{f^{(4)}(x)}{4!}h^4 - \frac{f^{(5)}(x)}{5!}h^5 + O(h^6)$$

$$f(x+h) - f(x-h) = 2f'(x)h + 2 \cdot \frac{f^{(3)}(x)}{3!}h^3 + 2 \cdot \frac{f^{(5)}(x)}{5!}h^5 + O(h^7)$$

$$D_h = f'(x) + \frac{f^{(3)}(x)}{3!}h^2 + \frac{f^{(5)}(x)}{5!}h^4 + O(h^6) \quad \text{--- (1)}$$

$$D_{\frac{4h}{3}} = f'(x) + \frac{f^{(3)}(x)}{3!} \left(\frac{4h}{3}\right)^2 + \frac{f^{(5)}(x)}{5!} \left(\frac{4h}{3}\right)^4 + O(h^6) \quad \text{--- (2)}$$

$$\frac{9}{16} \times (2) - (1) \Rightarrow$$

$$\frac{9}{16} D_{\frac{4h}{3}} - D_h = \cancel{\frac{9}{16} f'(x)} + \left(\frac{9}{16} - 1\right) f'(x) + \left(\frac{9}{16} - 1\right) \frac{f^{(3)}(x)}{5!} \left(\frac{4h}{3}\right)^4 + O(h^6)$$

$$\Rightarrow \frac{\frac{9}{16} D_{\frac{4h}{3}} - D_h}{\frac{9}{16} - 1} = f'(x) + \frac{f^{(5)}(x)}{5!} \left(\frac{4h}{3}\right)^4 + O(h^6) = D_h^1$$

$$\therefore D_h^1 = \frac{\frac{9}{16} D_{\frac{4h}{3}} - D_h}{\frac{9}{16} - 1}$$

Example:	x	1	1.2	1.4
	$f(x)$	1.2712	1.3642	1.5496

① Calculate $f'(1.2)$ using CD.

② Let, $f(x) = x \sin x + x^2 \cos x$, compute the upper bound error / maximum truncation error using CD, $E = [1, 1.4]$

⇒ ① $h = 0.2$

$$f'(1.2) = \frac{f(1.2+0.2) - f(1.2-0.2)}{2 \times 0.2} = \frac{1.5496 - 1.2712}{2 \times 0.2}$$

$$= 0.69475$$

② Using CD,

$$T. Error = \left| \frac{f'''(\xi)}{3!} h^2 \right| \quad f(x) =$$

$$f(x) = x \sin x + x^2 \cos x$$

$$f'(x) = \sin x + x \cos x + 2x \cos x - x^2 \sin x$$

$$= \sin x + 3x \cos x - x^2 \sin x$$

$$f''(x) = \cos x + 3 \cos x - 3x \sin x - 2x \sin x - x^2 \cos x$$

$$= 4 \cos x - 5x \sin x - x^2 \cos x$$

$$f'''(x) = -4 \sin x - 5 \sin x - 5x \cos x - 2x \cos x + x^2 \sin x$$

$$= -9 \sin x - 7x \cos x + x^2 \sin x$$

$$\left| \frac{f'''(\xi)}{3!} h^2 \right| = \left| \frac{(0.2)^2}{3!} \right| |f'''(\xi)|$$

$$= \left| \frac{(0.2)^2}{3!} \right| \left| -9 \sin(\xi_e) - 7\xi_e \cos(\xi_e) + \xi_e^2 \sin(\xi_e) \right|$$

$$= \left| \frac{(0.2)^2}{3!} \right| \left| 9 \sin(\xi_e) + 7(\xi_e) \cos(\xi_e) + \xi_e^2 \sin(\xi_e) \right|$$

$$= \left| \frac{(0.2)^2}{3!} \right| \left| 9 \sin(1.4) + 7(1.4) \cos(1) + (1.4)^2 \sin(1.4) \right|$$

Upper bound = 0.1073
(Answer)

$$\left| \frac{f^{(3)}(\xi)}{3!} \right|$$

$$\cos^3 x - \sin^2 x^2 - \sin^2 x^2 - \cos^2 x - \cos^2 x = (x)^3$$

$$\cos^3 x - \sin^2 x^2 - \cos^2 x =$$

$$\sin^2 x + \cos^2 x - \cos^2 x^2 - \sin^2 x - \sin^2 x = (x)^3$$

$$\sin^2 x + \cos^2 x - \sin^2 x =$$

$$\left| \frac{f^{(3)}(\xi)}{3!} \right| \quad \left| \frac{f^{(3)}(\xi)}{3!} \right| \quad \left| \frac{f^{(3)}(\xi)}{3!} \right|$$